Scientific Reports

https://doi.org/10.1038/s41598-026-66692-x

Article in Press

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Received: 25 May 2026

Accepted: 10 August 2026

Published online: 11 August 2026

Cite this article as: Ferdous M., Utso M.Z., Gani R. *et al.* An explainable graph based inception framework for *Aegle marmelos* leaf disease classification. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-66692-x

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An Explainable Graph Based Inception Framework for Aegle marmelos Leaf Disease Classification

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ABSTRACT
Aegle marmelos (bael) is a crucial medicinal tropical crop that has been largely overlooked in computational plant pathology studies. This study introduces AMnet, a cutting-edge deep learning framework that combines an InceptionV3 backbone with a fixed graph convolutional network. This integration effectively captures both within-region disease texture and between-region spatial propagation patterns, enabling the automated four-class classification of Aegle marmelos leaf diseases: Cercospora, healthy leaf, leaf curl, and leaf spot. The graph module constructs a 25-node spatial graph directly from the CNN feature maps, enabling end-to-end training without external preprocessing. Compared with MobileNetV2, InceptionV3, VGG19, and DenseNet201, AMnet demonstrated superior performance, achieving 98.83% accuracy, 98.84% F1-score, 99.95% PR-AUC, and 98.45% MCC, along with the fastest inference time of 1.83 ms. Its robustness was validated using bootstrap confidence interval estimation and four-fold cross-validation. PCA-based clustering analysis, supported by silhouette scoring and Davies–Bouldin indexing, confirmed clear class separability in the learned embeddings. Grad-CAM and LIME visualizations verified that the predictions were based on biologically significant leaf regions rather than background artifacts. A Gradio-based prototype demonstrated its practical deployment potential. Although broader field validation is still required, AMnet is an accurate, interpretable, and reproducible framework for diagnosing Aegle marmelos leaf disease.

Keywords: Aegle marmelos leaf disease classification; AMnet, InceptionV3, fixed graph convolutional network, deep learning; plant disease diagnosis, explainable AI, precision agriculture

1 Introduction

Plant diseases remain a major challenge in agricultural production because they reduce crop quality, destabilize yield, and increase management costs. Therefore, early and accurate disease identification is essential for timely intervention and sustainable crop protection. In many agricultural settings, diagnosis still relies on visual inspection by farmers and plant specialists. Although expert assessments are valuable, they are often time-consuming, subjective, and difficult to apply consistently across large cultivation areas.

Aegle marmelos, commonly known as bael, is a medicinal and fruit-bearing plant of substantial nutritional, therapeutic, and economic importance. The leaves are affected by diseases that produce visible changes in color, texture, venation, shape, and surface morphology. This study considers four categories: Cercospora leaf, healthy leaf, leaf curl, and leaf spot. These categories exhibit symptoms such as chlorosis, discoloration, deformation, and irregular lesion patterns.

However, similarities among their visual characteristics, particularly during early or intermediate disease stages, make both manual and automated classification challenging.

Deep learning has demonstrated considerable potential for image-based plant disease recognition by automatically learning discriminative features from leaf images [1,2]. Convolutional neural networks (CNNs) can identify disease-related patterns such as lesion shape, colour variation, texture irregularity, and structural deformation. Architectures such as MobileNetV2, InceptionV3, VGG19, and DenseNet201 have therefore been widely adopted in agricultural image classification. Nevertheless, conventional CNNs primarily learn features through local convolutional operations and do not explicitly represent relationships among spatially separated disease regions. Moreover, high classification accuracy alone does not guarantee reliable deployment, as a model may rely on background artifacts or dataset-specific visual cues rather than biologically meaningful symptoms [3].

Recent studies on plant diseases have incorporated attention mechanisms, graph neural networks, semi-supervised learning, and explainable artificial intelligence to improve feature representation and model transparency [4–6]. Graph-based methods are particularly useful for modeling relational dependencies among image regions. Existing CNN–graph frameworks for plant disease analysis, however, commonly employ complex multi-module architectures, represent relationships among individual leaves, or construct graphs via superpixel segmentation, edge augmentation, or other preprocessing steps. Such approaches may increase computational complexity and make graph construction dependent on segmentation quality or externally defined structural information. Furthermore, relatively little attention has been paid to Aegle marmelos, leaving an important tropical medicinal crop underrepresented in computational plant pathology.

To address these limitations, this study proposes AMnet, an end-to-end graph-augmented Inception framework for four-class classification of Aegle marmelos leaf diseases. Unlike conventional CNN models that process spatial features without explicit relational reasoning, AMnet converts an InceptionV3 feature map into a 5 × 5 grid of 25 graph nodes using adaptive average pooling. A fixed eight-neighbor adjacency structure with self-connections is then used to model the dependencies among neighboring leaf regions. This deterministic graph construction differs from graph-based plant disease methods that rely on superpixels, externally generated graphs, and relationships among multiple leaf instances. The graph-enhanced features were combined with the original CNN representation via residual and feature-fusion operations, enabling the model to preserve local disease textures while learning region-level spatial interactions. Consequently, AMnet provides a comparatively simple and reproducible CNN–GCN architecture that requires no external graph-generation pipeline and can be trained directly on leaf images. The main contributions of this study are as follows.

- **End-to-end CNN–GCN architecture with direct spatial graph construction:** AMnet integrates an InceptionV3 backbone with a fixed graph convolutional module that operates on a 5 × 5 grid derived directly from CNN feature maps. In contrast to standalone CNNs and graph-based methods that require superpixel segmentation or external structural priors, AMnet jointly learns local visual characteristics and spatial relationships among neighboring disease regions without additional graph preprocessing.
- **A domain-specific framework for an underrepresented medicinal crop:** This study provides a systematic investigation of automated disease classification for Aegle marmelos and evaluates AMnet against established CNN architectures under a common experimental protocol. To the best of our knowledge, it is the first study to apply graph-based spatial reasoning to the classification of Cercospora leaf, healthy leaf, leaf curl, and leaf spot in this crop.
- **Trustworthy, explainable, and practically deployable evaluation:** The proposed framework is evaluated using accuracy, precision, recall, F1-score, PR-AUC, MCC,

confidence intervals, and four-fold cross-validation. PCA-based clustering, silhouette scoring, Davies–Bouldin indexing, and centroid-distance analysis are used to assess the separability of the learned representations. Grad-CAM and LIME are further employed to assess whether predictions rely on biologically meaningful leaf regions, while a Gradio-based prototype demonstrates AMnet's practical potential for rapid, explainable disease diagnosis.

The remainder of this paper is organized as follows. **Section 2** reviews relevant deep learning, graph-based learning, semi-supervised learning, and plant disease classification studies. **Section 3** describes the dataset, preprocessing procedure, baseline architectures, proposed AMnet framework, feature-space analysis, and explainability methods. **Section 4** presents the experimental results and validation analyses. **Section 5** discusses the principal findings, practical implications, limitations, and future directions. Finally, **Section 6** concludes the study.

2 Related Works

2.1 Deep Neural Network-Based Plant Disease Diagnosis and Explainable AI

Deep neural networks have substantially improved image classification by learning discriminative features directly from visual data. Attention-based architectures further enhance this capability by emphasizing relevant regions while suppressing irrelevant information. Kanumuri et al. [7] integrated CBAM with ResNet152V2 for pneumonia classification, while Gao et al. [8] combined CNN-based feature extraction with Transformer attention in X-Profiler. Other hybrid architectures, including CNN–BiGRU–Transformer [9], SkinProNet [10], ESADL [11], C-Swin [12], ViT-based self-attention [13], and tri-domain attention [14], have demonstrated the effectiveness of attention mechanisms in complex image classification tasks. These developments provide transferable strategies for identifying subtle colour, texture, and structural variations in diseased leaves.

In plant disease diagnosis, CNN, Transformer, and attention-based frameworks have achieved strong classification performance. Aboelenin et al. [15] combined CNN architectures with Vision Transformers for apple and corn leaf disease recognition, achieving accuracies of 99.24% and 98%, respectively. D, T. L. N., and Prusty [16] proposed ALNet, a lightweight attention-based CNN for grapevine disease classification, reporting up to 100% accuracy with only 0.17 million parameters. Sandhi et al. [17] developed a multi-architecture framework integrating ResNet, DenseNet, EfficientNet, and MBConv-SE blocks, achieving accuracies of 99.99%, 99.60%, and 99.68% on apple, guava, and mango datasets, respectively. Sandhi et al. [18] subsequently introduced SAAEffNet, which combines CycleGAN-based synthetic augmentation, EfficientNetB3, and CBAM attention for mango disease classification, achieving 98.40% accuracy, precision, and recall and a 98.39% F1-score.

Data-efficient learning strategies have also been investigated to reduce reliance on large manually annotated datasets. Ilsever and Baz [19] proposed a ResNet-50 Mean-Teacher framework for plant disease recognition, improving accuracy by up to 5.52% under limited-label conditions. Pham, Nguyen, and Nguyen [20] developed an ambiguity-aware semi-supervised framework for coffee and banana leaf disease classification, reporting precision values of 99.46% and 100.0% using only 50% labelled data. Related studies have applied strong-teacher consistency [21], pseudo-label correction [22], masked autoencoder reconstruction [23], and semi-supervised masked autoencoding [24] to improve learning from limited annotations. Islam et al. [25] also compared supervised, self-supervised, and semi-supervised strategies, demonstrating the broader potential of representation learning when labeled samples are scarce.

Explainable artificial intelligence is particularly important in plant disease diagnosis because high predictive accuracy alone does not confirm that a model uses biologically meaningful symptoms. Singh, Sharma, and Dey [26] combined CBAM-VGG16 with Grad-CAM, Grad-CAM++, and layer-wise relevance propagation, achieving up to 98.87% accuracy while visualizing the image regions responsible for predictions. Sandhi et al. [27] proposed a modified ResNet101 framework with dual-stream feature extraction and a hybrid genetic algorithm–particle swarm optimization for pomegranate disease classification. Their method achieved 99.10% accuracy under five-fold cross-validation and employed Grad-CAM++ to localize disease-affected regions. Similarly, Sanyal et al. [28] used soft labeling and pseudo-labeling in Pseudo CNN, achieving 96.4% accuracy.

These studies demonstrate substantial progress in CNN-based feature extraction, attention-guided learning, synthetic augmentation, semi-supervised training, and visual explanation. However, most existing work focuses on common fruit or crop species, while medicinal plants remain comparatively underrepresented. Moreover, explainability is frequently presented only through qualitative heatmaps, with limited integration of feature-space analysis, statistical reliability assessment, and practical deployment. These limitations motivate the development of an accurate and explainable framework specifically designed for diagnosing Aegle marmelos leaf disease.

2.2 Graph Convolutional Networks and CNN–GCN Frameworks

Graph convolutional networks extend conventional deep learning by explicitly modeling relationships among nodes or feature regions. Xu and Zhang [29] proposed GCNAA for digital ink Chinese character recognition by representing sampling points as graph vertices, achieving 98.7% accuracy. Yan et al. [30] introduced RDD-GCN, which uses reinforced dependency graphs and denoising mechanisms for sarcasm detection. Thekumparampil et al. [31] developed AGNN for graph-structured semi-supervised learning, replacing conventional propagation layers with graph attention mechanisms. These studies demonstrate that graph-based learning can capture relational and contextual information that may not be adequately represented through independent feature processing.

Graph convolutional and attention mechanisms have also been widely combined for spatial–temporal modelling. Yang et al. [32] developed AT-GCN by integrating GCNs, attention mechanisms, and GRUs for traffic prediction. Su, Huang, and Li [33] proposed an adaptive graph convolutional framework with spatial and frequency-domain attention, while Luo, Zhu, Zhang, and Li [34] introduced a dynamic GCN with attention fusion that outperformed 18 baseline methods across four traffic datasets. Yang et al. [35] further proposed TARGCN, which uses node-embedded GCNs, GRUs, and temporal attention. Beyond transportation, Chang, Li, and Duan [36] developed GANM for fake news detection, using graph-structured propagation and memory-enhanced global attention. Although these methods demonstrate the effectiveness of graph learning, their application is often limited by the quality of graph construction, computational complexity, scalability, and sensitivity to noisy relationships.

In plant disease recognition, CNN–GCN frameworks have been developed to combine visual feature extraction with relational reasoning. Surekha and Rao [37] proposed AdjLeafGNN, which integrates CNN, ASPP, CSAM, and GNN modules for leaf disease classification and adjacent-leaf disease spread prediction on the PlantVillage dataset. The framework achieved 98.88% accuracy and a 98.71% F1-score but relied on a relatively complex multi-module architecture and relationships among separate leaf instances. Sundhar et al. [38] introduced a hybrid GAT–GCN framework that uses superpixel segmentation and edge augmentation for the classification of apple, potato, and sugarcane diseases. Although the method demonstrated strong performance, its

graph construction depends on an external segmentation process and the quality of the generated superpixels and edges.

Existing CNN–GCN plant disease methods therefore confirm the value of combining convolutional representations with graph-based reasoning. Nevertheless, many approaches require superpixel segmentation, externally defined graph structures, edge augmentation, or complex combinations of multiple modules. Such graph-generation procedures may increase preprocessing requirements and introduce sensitivity to segmentation or graph quality. In addition, limited research has examined graph-based reasoning for medicinal plants or integrated graph learning with statistical validation, complementary explainability methods, feature-space evaluation, and prototype deployment. **Table 1** shows the overall study summaries.

Table 1. Comparative Analysis of Existing Deep Learning, Graph-Based, and Semi-Supervised Approaches

Ref	Dataset	Method	Model	Accuracy	Limitation
[12]	IQ-OTH/NCCD CT Dataset	CNN-Transformer lung cancer classification	C-Swin	96.26%	Small dataset and overfitting risk
[14]	GTZAN	Music genre classification	CNN–Transformer + Tri-domain Attention	96.96%	Single benchmark dataset dependency
[23]	Medical Imaging Datasets	Pseudo-label refinement	ViT + MAE + Pseudo-labeling	Improved low-label performance	Sensitivity to pseudo-label noise
[24]	Three Medical Segmentation Datasets	Semi-supervised segmentation	SEMI-PLC	Improved Dice and 95HD	Increased training complexity
[29]	CASIA, SCUT-COUCH2009, HIT-OR3C	Graph-based handwritten character recognition	GCNAA	98.7%	Sensitivity to graph construction quality
[31]	Citation Network Datasets	Semi-supervised graph learning	AGNN	Outperformed competing methods	Limited scalability analysis
[34]	Four Traffic Datasets	Dynamic graph traffic prediction	Dynamic GCN + Attention Fusion	Outperformed 18 baselines	High computational complexity
[36]	Social Media Propagation Datasets	Fake news detection	GANM	Promising detection performance	Sensitivity to noisy interactions

[38]	Apple, Potato, Sugarcane Datasets	Hybrid graph-based leaf classification	GAT-GCN	F1: 0.9818, 0.9743, 0.8799	Lower performance on the sugarcane dataset

AMnet addresses these gaps by constructing a fixed spatial graph directly from an InceptionV3 feature map. Thus, the contribution of AMnet lies in its direct and reproducible CNN-to-graph construction, its application to an underrepresented medicinal crop, and its integration with explainability, statistical reliability analysis, feature-space validation, and practical deployment.

3 Materials and Methods

Fig. 1 illustrates the overall experimental workflow of the proposed AMnet framework for Aegle marmelos leaf disease classification. The pipeline begins with dataset acquisition, preprocessing, normalization, and augmentation to improve data diversity and model generalization. Multiple supervised baseline models, including MobileNetV2, InceptionV3, VGG19, and DenseNet201, are trained and compared with the proposed AMnet architecture. AMnet integrates an InceptionV3 backbone with a Fixed-GCN module to capture both deep visual features and spatial relationships among leaf regions. The learned representations are further evaluated using classification metrics, explainable AI techniques, PCA-based clustering analysis, and statistical validation methods. Finally, a Gradio-based prototype application is developed to demonstrate the proposed system's practical deployment capabilities for real-time, explainable leaf disease diagnosis.

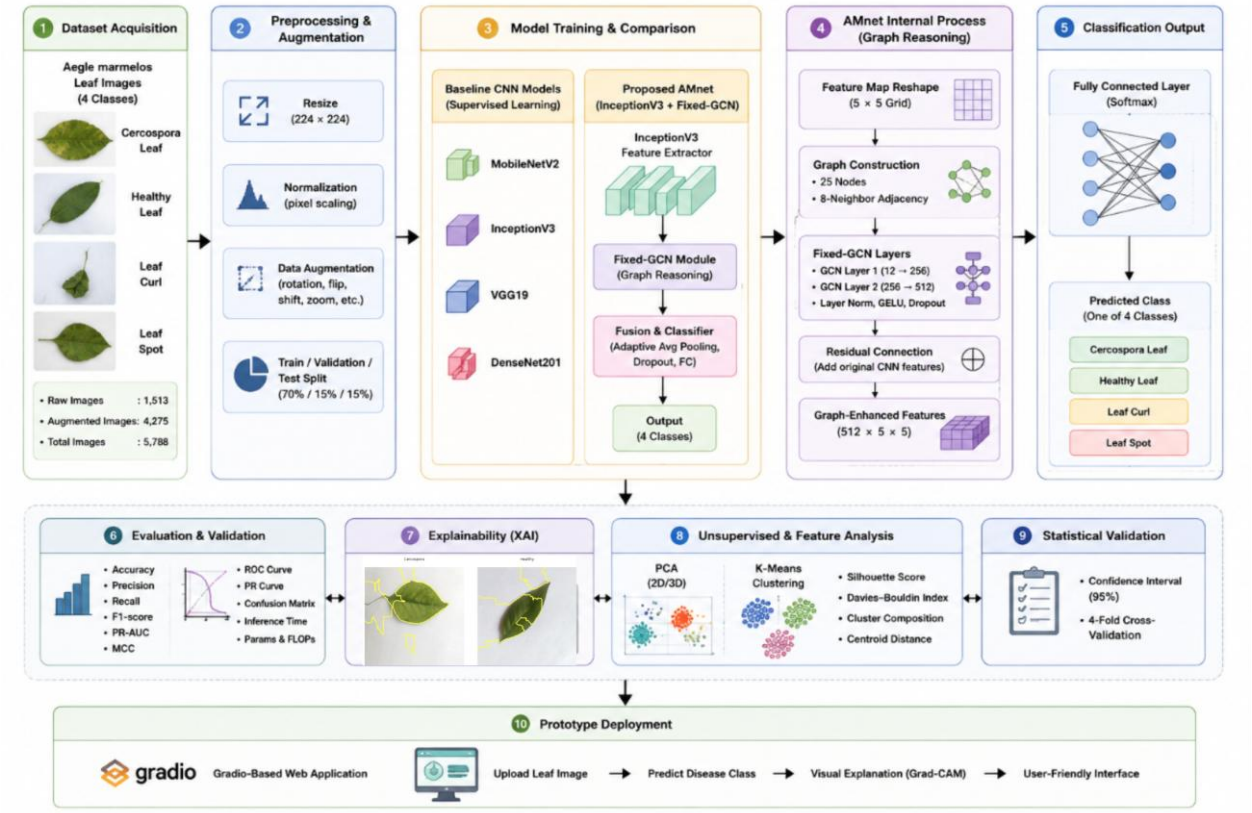


Fig. 1. Overall Working Procedure

3.1 Dataset Description

This study uses a publicly available smartphone image dataset for automated Aegle marmelos leaf disease classification [39]. The selected subset contains four categories: Cercospora leaf, healthy

leaf, leaf curl, and leaf spot. Diseased leaves exhibit symptoms such as chlorosis, discoloration, deformation, irregular lesions, and texture variation, whereas healthy leaves generally show normal structure and uniform green pigmentation.

3.1.1 Image Acquisition

The images were collected during December 2024 from different locations in Rangpur, Bangladesh. Leaf samples were photographed under natural conditions using a Xiaomi POCO X3 Pro smartphone equipped with a 48-MP primary camera. Five images were captured from different viewing angles for each leaf to represent variations in orientation, texture, shape, discoloration, and disease appearance. A white background was used to reduce visual noise and improve the visibility of symptomatic regions. The images were labelled according to plant type and disease condition. However, the dataset does not consider illumination intensity, weather conditions, inter-annotator agreement, or disease-severity stages. **Fig. 2** presents representative samples from the four classes and illustrates the main visual differences considered during model training and evaluation.

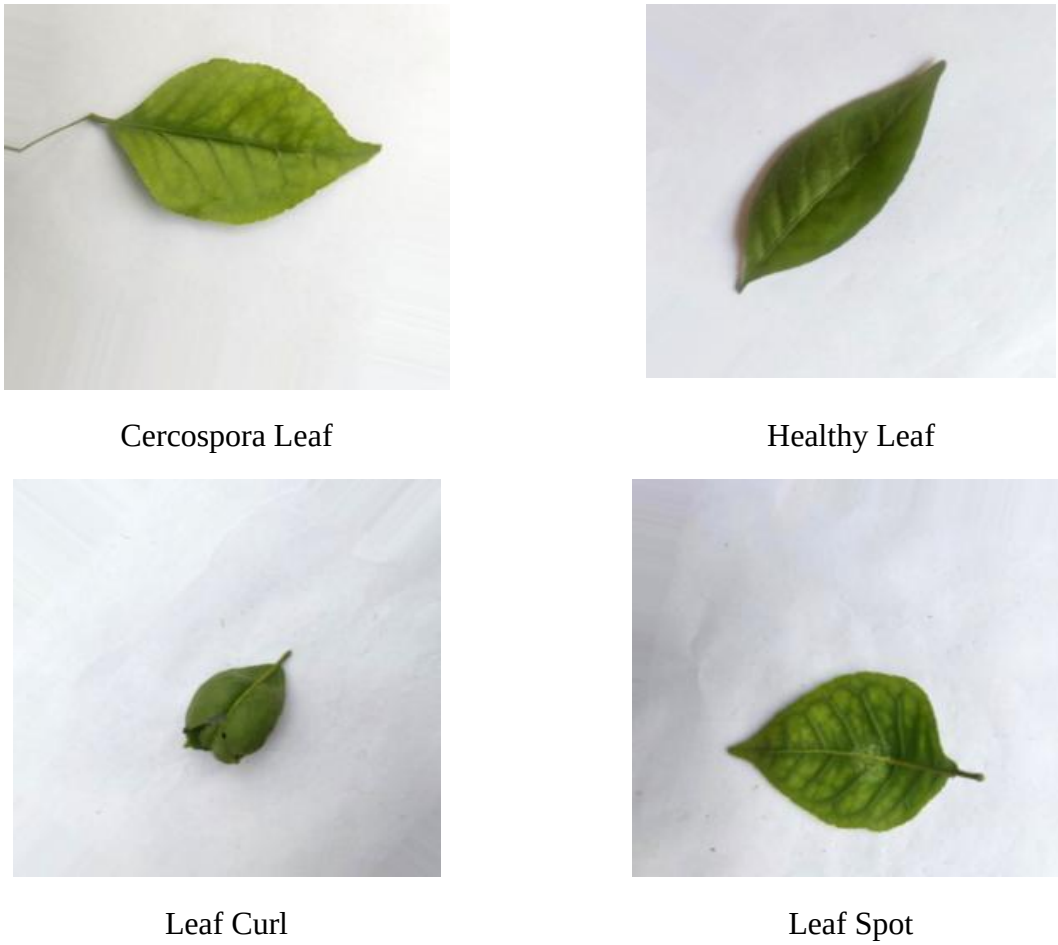


Fig. 2. Dataset Sample Images

3.1.2 Dataset Preparation and Class Distribution

The dataset contains 1,513 raw JPG images with a resolution of 700 × 700 pixels. Data augmentation was used to increase image diversity through rotation, horizontal flipping, width

and height shifting, shearing, and zooming. The augmented images were subsequently resized to 224 × 224 pixels for model input. The augmentation process increased the dataset to 4,275 images.

To prevent data leakage, the training, validation, and test partitions were created from the raw images before augmentation. Augmentation was then applied exclusively to the training partition, ensuring that augmented versions of validation or test images were not included in training. This procedure supports a more reliable assessment of model generalization to unseen samples.

Table 2 summarizes the class-wise distribution of raw and augmented Aegle marmelos leaf images used in this study. The dataset contains 1,513 raw images, which were expanded to 4,275 augmented images through data augmentation. The four classes are relatively balanced, with raw image counts ranging from 370 images for Leaf Spot to 386 images for Cercospora. A similar balance is maintained after augmentation, where each class contains more than 1,000 images. This balanced distribution helps reduce class bias during model training and supports fair evaluation of classification performance across healthy and diseased leaf categories.

Table 2. Class-wise distribution of the Aegle marmelos leaf dataset

Class	Raw Images	Augmented Images
Leaf Spot	370	1046
Leaf Curl	380	1067
Cercospora	386	1110
Healthy	377	1052
Total	**1,513**	**4,275**

3.2 Baseline Models Architecture

This section describes the architectural characteristics of the baseline deep learning models used for comparative evaluation, including MobileNetV2, InceptionV3, VGG19, and DenseNet201.

3.2.1 MobileNetV2 Architecture

MobileNetV2 was used as a lightweight convolutional baseline model for leaf disease classification. The architecture is based on inverted residual blocks, depthwise separable convolutions, and linear bottleneck layers, which reduce computational cost while preserving useful spatial features. The pretrained feature extraction layers were used to extract leaf texture, shape, and color-related patterns. The original classification head was replaced with a task-specific fully connected layer followed by softmax activation for four-class classification.

3.2.2 InceptionV3 Architecture

InceptionV3 was employed as a multi-scale feature extraction model to capture disease patterns at different receptive-field sizes. The architecture uses inception modules with factorized convolutions, allowing the network to learn both local and global visual features efficiently. In this study, the pretrained InceptionV3 backbone was used to extract discriminative leaf representations from the input images. Its final classification layer was modified to produce four output classes corresponding to the target leaf categories.

3.2.3 VGG19 Architecture

VGG19 was selected as a deep convolutional baseline with a simple and uniform architecture. It consists of sequential convolutional layers using small 3 × 3 kernels, followed by max-pooling layers to progressively reduce spatial dimensions while increasing feature depth. This structure enables the model to learn hierarchical visual patterns from low-level edges to high-level disease-

specific textures. The original classifier was replaced with a four-class output layer for *Aegle marmelos* leaf disease classification.

3.2.4 DenseNet201 Architecture

DenseNet201 was used as a strong supervised baseline because of its dense connectivity between convolutional layers. In this architecture, each layer receives feature maps from all preceding layers, which encourages feature reuse and improves gradient flow during training. This design helps the model capture fine-grained disease symptoms such as spots, curling, discoloration, and texture variation. The pretrained DenseNet201 classifier was adapted by replacing the final layer with a four-neuron softmax classifier for the target classes.

3.3 AMnet Model Architecture

AMnet integrates an InceptionV3 backbone with a fixed graph convolutional network to jointly learn local visual patterns and spatial relationships among different leaf regions. As illustrated in **Fig. 3**, each input image is resized to 224 × 224 × 3 and passed through the pretrained InceptionV3 backbone up to the Mixed_7c block. The resulting feature map contains 2048 channels and represents high-level disease characteristics such as lesions, discoloration, texture variation, venation, and structural deformation.

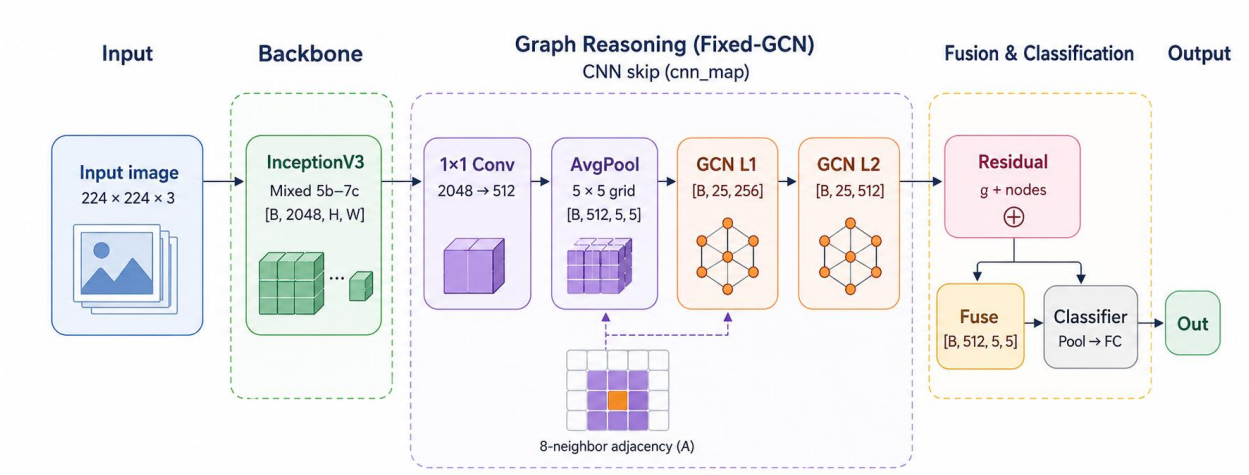


Fig. 3. AMnet Architecture

A 1 × 1 convolutional projection reduces the channel dimension from 2048 to 512, followed by batch normalization and GELU activation. Adaptive average pooling then transforms the projected feature map into a fixed 5 × 5 spatial representations. The operation divides the incoming feature map into 25 spatial bins and calculates the average activation within each bin. Consequently, the pooled feature tensor has dimensions 512 × 5 × 5, regardless of the original spatial dimensions of the backbone output. The pooled tensor is reshaped into a node-feature matrix,

$$X \in R^{25 \times 512} \tag{1}$$

where each of the 25 nodes represents one region of the 5 × 5 feature grid, and each node contains a 512-dimensional CNN feature vector. This direct transformation allows graph construction from learned CNN representations without superpixel segmentation or external preprocessing.

A fixed 8-neighbor graph is constructed according to the spatial positions of the nodes in the grid. Each node is connected to its horizontal, vertical, and diagonal neighbors. Corner and boundary nodes have fewer neighbors because some surrounding positions fall outside the grid. Self-

connections are also included to preserve each node's original information. For nodes i and j, located at grid coordinates (r_i,c_i) and (r_j,c_j), respectively, the adjacency matrix $A \in R^{25\times25}$ is defined as,

$$A_{ij} = \begin{cases} 1, & |r_i - r_j| \leq 1 \text{ and } |c_i - c_j| \leq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

The adjacency matrix is symmetrically normalized as,

$$\hat{A} = D^{-1/2} A D^{-1/2} \quad (3)$$

where D is the diagonal degree matrix. This normalization balances the contribution of nodes with different numbers of neighbors and supports stable graph propagation.

The node features are processed through two fixed-GCN layers. The first layer transforms the node representation from 512 to 256 dimensions, while the second restores it from 256 to 512 dimensions:

$$\begin{aligned} H^{(1)} &= \mathrm{GELU}\left(\mathrm{LN}\left(\hat{A}XW^{(1)}\right)\right), \\ H^{(2)} &= \mathrm{GELU}\left(\mathrm{LN}\left(\hat{A}W^{(1)}W^{(2)}\right)\right) \end{aligned} \quad (4)$$

where $W^{(1)} \in R^{512\times256}$, $W^{(2)} \in R^{256\times512}$ and LN denotes layer normalization. Dropout is applied within the GCN layers to reduce overfitting. The resulting graph representation has dimensions,

$$H^{(2)} \in R^{25\times512} \quad (5)$$

Because the original CNN node matrix and final GCN output have identical dimensions, a residual connection is implemented through element-wise addition:

$$X_{\mathrm{res}} = X + H^{(2)} \quad (6)$$

This operation preserves the original regional CNN information while incorporating spatial context learned through graph message passing. The residual node representation is then reshaped into a 512 × 5 × 5 spatial feature maps.

For CNN–GCN feature fusion, the graph-enhanced feature map is concatenated channel-wise with the original pooled CNN feature map. This produces a fused tensor of dimensions 1024 × 5 × 5. A fusion block consisting of a 1 × 1 convolution, batch normalization, and GELU activation reduces the channel dimension from 1024 to 512. Finally, global adaptive average pooling converts the fused feature map into a 512-dimensional vector. Dropout is applied before a fully connected layer generates four class logits corresponding to Cercospora leaf, healthy leaf, leaf curl, and leaf spot.

The fixed spatial graph is identical for all input images, while the node features are learned independently from each image. Therefore, AMnet remains end-to-end trainable and requires no external graph-generation procedure. By combining CNN-derived local appearance features with graph-refined regional context, the framework captures both disease-specific visual patterns and spatial dependencies among neighboring leaf regions.

Algorithm 1 summarizes the complete computational workflow of AMnet, beginning with input image preprocessing and InceptionV3-based feature extraction, followed by channel reduction, construction of a fixed 8-neighbor graph, two-layer GCN reasoning, CNN-GCN feature fusion, and final classification. The algorithm shows how the model transforms a leaf image into spatial graph nodes and applies graph-based message passing to capture local relationships among neighboring leaf regions. This design allows AMnet to combine the strong visual representation capabilities of InceptionV3 with the structured spatial reasoning of the Fixed-GCN module. As a result, the model can learn both disease-specific visual patterns and region-level dependencies, supporting more reliable *Aegle marmelos* leaf disease classification.

Algorithm 1 AMnet: InceptionV3 with Fixed-GCN Reasoning

Input: Leaf image
Output: Predicted class label $\hat{y}$ and class probabilities
Input: Leaf image $I \in R^{224\times224\times3}$
Output: Predicted class label $\hat{y}$.

1. Receive the input leaf image I.
2. Resize and normalize the image for InceptionV3 input.
3. Extract deep convolutional feature maps using the pretrained InceptionV3 backbone up to the Mixed_7c block.
4. Apply a 1 × 1 convolution to reduce the feature dimension from 2048 to 512.
5. Apply adaptive average pooling to generate a 5×5 spatial feature grid.
6. Convert the 5 × 5 feature grid into 25 graph nodes.
7. Construct a fixed 8-neighbor adjacency matrix with self-loops for the graph nodes.
8. Normalize the adjacency matrix using symmetric degree normalization.
9. Pass the node features through the first Fixed-GCN layer to obtain 256-dimensional graph features.
10. Pass the output through the second Fixed-GCN layer to restore 512-dimensional graph features.
11. Add the original CNN node features to the GCN output using a residual connection.
12. Reshape the graph-enhanced node features back into a spatial feature map.
13. Concatenate the original CNN feature map and the graph-enhanced feature map.
14. Apply the fusion block using 1×1 convolution, batch normalization, and GELU activation.
15. Apply adaptive average pooling and dropout.
16. Pass the fused feature representation through the fully connected classifier.
17. Generate the final predicted class label $\hat{y}$

Return: Predicted class label $\hat{y}$ and class probabilities.

3.4 Experimental Setup

Table 3 summarizes the principal hyperparameters used to train the classification model. All input images were resized to 224×224 pixels, and the output layer contained four neurons corresponding to the four target classes. Cross-entropy loss was minimized using the Adam optimizer with a learning rate of 1×10e−4. Training was conducted with a batch size of 32, while early stopping with a patience of five epochs was applied to reduce overfitting by terminating training when validation performance no longer improved.

Table 3. Training Configuration and Hyperparameter Settings of the Proposed Model

Category	Hyperparameter	Value
Input	Image size	(224 × 224)
Classification	Output classes	4
Optimization	Loss	Cross-Entropy
	Optimizer	Adam
	Learning rate	$1\times10e^{-4}$
Training	Batch size	32
	Early stopping patience	5

3.5 Unsupervised Learning-Based Feature Analysis

This study employs unsupervised learning to examine the separability of AMnet feature embeddings without relying on class labels during clustering. The analysis evaluates whether the learned deep features naturally form meaningful groups corresponding to the four-leaf classes.

The elbow method identifies the suitable number of clusters by measuring the change in inertia across different K values. The silhouette score assesses cluster compactness and separation, while the Davies–Bouldin index provides an additional measure of clustering quality, where lower values indicate better separation. After determining the appropriate number of clusters, K-means clustering groups the AMnet feature embeddings into distinct clusters. Finally, PCA reduces the high-dimensional embeddings into a two-dimensional space to visualize class-wise distribution, overlap, and separation.

3.6 Explainable AI (XAI)

This study incorporates explainable AI to interpret AMnet's predictive behavior and assess whether the model focuses on disease-relevant leaf regions. Grad-CAM generates class-discriminative heatmaps by highlighting important image regions that contribute strongly to the final prediction, such as lesions, discoloration, vein deformation, and texture changes.

LIME provides a complementary local explanation by perturbing image segments and identifying the regions that most influence the model’s decision. Together, Grad-CAM and LIME enhance model transparency by showing whether AMnet bases its predictions on meaningful leaf symptoms rather than background or irrelevant visual patterns.

4 Results and Analysis

4.1 Performance Analysis

To determine the contribution of graph-based learning, AMnet was compared directly with the InceptionV3 baseline. As shown in **Table 4**, adding the GCN markedly improved accuracy from 68.22% to 98.83%, F1 score from 68.88% to 98.84%, PR AUC from 80.96% to 99.95%, and MCC from 59.14% to 98.45%. This improvement can be attributed to the complementary roles of the two components. InceptionV3 extracts discriminative visual features from the input images, whereas the GCN models relationships and dependencies among those features. By incorporating this structural information, AMnet can better separate visually similar classes, reduce ambiguous predictions, and preserve class-specific patterns. Consequently, the combined architecture produces more accurate, balanced, and reliable classifications than InceptionV3 alone. This explains its consistent gains across metrics.

Table 4. Performance comparison of the models

Model Name	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	PR AUC (%)	MCC (%)
MobileNetV2	92.06	92.05	92.13	92.01	97.19	89.46
InceptionV3	68.22	75.46	67.56	68.88	80.96	59.14
VGG19	98.13	98.14	98.17	98.15	99.84	97.51
DenseNet201	96.26	96.35	96.21	96.23	99.15	95.04
ResNet50	98.36	98.40	98.37	98.37	99.88	97.83
AMnet	**98.83**	**98.86**	**98.85**	**98.84**	**99.95**	**98.45**

Beyond overall classification accuracy, the reliability of the predictions was further examined using specificity and agreement measures. The results presented in **Table 5** show an average specificity of 99.38%, indicating that the model rarely misclassified negative samples. In addition, Cohen's kappa reached 97.51%, reflecting near-perfect agreement between the predicted and actual labels. These outcomes further confirm the model's strong consistency and dependable classification performance.

Table 5. Specificity and agreement performance of AMnet

Average Specificity (%)	Cohen's Kappa (%)
99.38	97.51

4.2 Computation Analysis

In addition to predictive performance, computational efficiency is important for determining whether a model can support real-time applications. As presented in **Table 6**, AMnet achieved the shortest inference time of 1.83 ms and the highest processing speed of approximately 547.89 frames per second. Although MobileNetV2 required the fewest parameters, FLOPs, and storage space, its inference speed was considerably lower than that of AMnet. The remaining models required longer inference times and processed fewer frames per second. These findings demonstrate that AMnet offers superior inference efficiency despite its relatively large parameter count and model size, making it suitable for applications requiring rapid and reliable predictions.

Table 6. Parameter summary of the models

Model Name	Inference Time (ms)	Frames/sec (approx.)	FLOPs (G)	No. of Params	Size (MB)
MobileNetV2	4.39	227.74	0.613	2, 273, 358	9.08
InceptionV3	5.06	197.55	5.693	21, 827, 374	84.05
VGG19	8.99	111.19	39.038	20, 030, 542	76.48
DenseNet201	7.64	130.84	8.632	18, 345, 038	71.72
ResNet50	5.15	194.20	7.751	23,612,302	90.53
AMnet	**1.83**	**547.89**	**5.764**	**23,626,212**	**90.13**

4.3 Loss Curve Analysis

To further examine the learning behaviour of the evaluated models, the training and validation loss curves are presented in **Fig. 4**. MobileNetV2 showed a steady reduction in training loss, while its validation loss stabilized after approximately 12 epochs, suggesting limited further improvement. InceptionV3 displayed noticeable validation-loss fluctuations, indicating less stable convergence. In contrast, VGG19, DenseNet201, and ResNet50 achieved smoother convergence with relatively small gaps between the two curves. AMnet reduced its loss rapidly within eight epochs, although the validation loss levelled off after the fourth epoch. Overall, the curves indicate

that AMnet learned efficiently, while ResNet50 and VGG19 demonstrated the most stable long-term convergence.

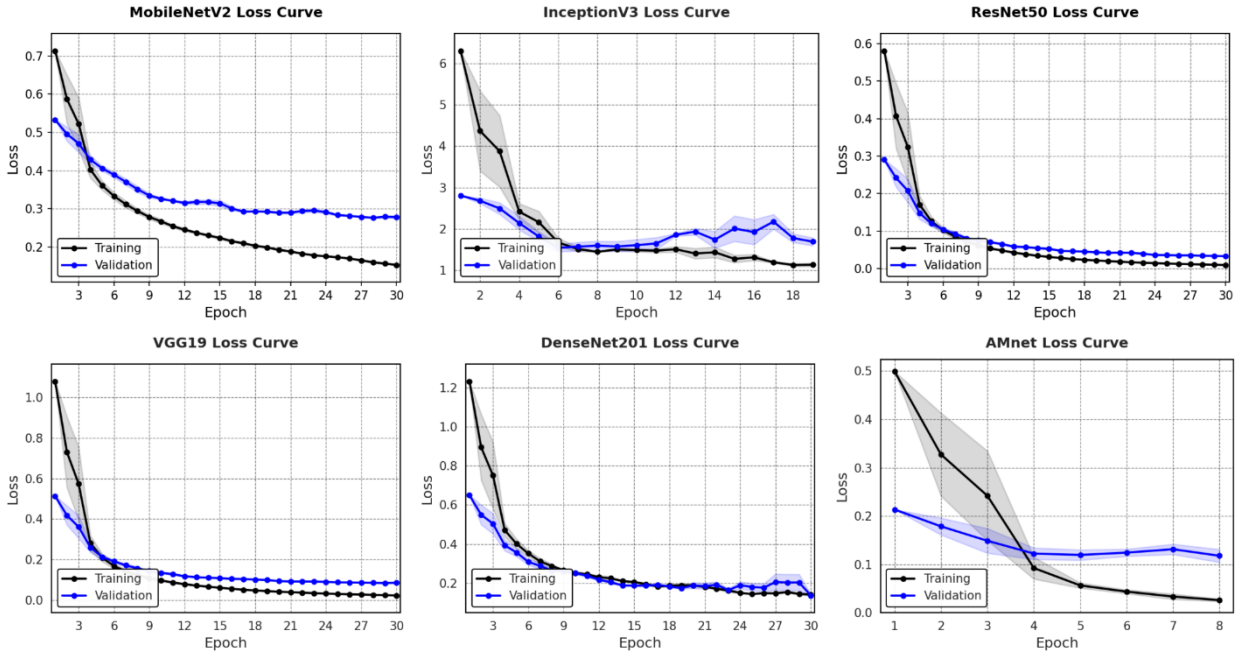


Fig. 4. Models Loss Curves

4.4 Confusion Matrix Analysis

As shown in Fig. 5, InceptionV3 exhibited substantial confusion among visually similar classes, particularly between leaf-curl, leaf-spot, and healthy leaves. In contrast, AMnet achieved balanced class-wise accuracy ranging from 97.1% to 99.1%, with only a few misclassifications. This improvement is likely due to the GCN component, which captures relationships among extracted visual features and helps distinguish disease classes with similar colour, texture, and shape characteristics.



Fig. 5. Models Confusion Matrix

4.5 ROC and PR Curve Analysis

The discriminative ability of the evaluated models was further examined using the ROC curves in **Fig. 6**. Most models produced curves close to the upper-left corner, indicating strong class separation and low false-positive rates. VGG19, ResNet50, DenseNet201, and AMnet achieved near-perfect AUC values, whereas InceptionV3 showed comparatively lower performance across the four classes. These differences suggest that the stronger models learned more distinctive class-specific features, allowing them to separate healthy and diseased leaves more effectively.

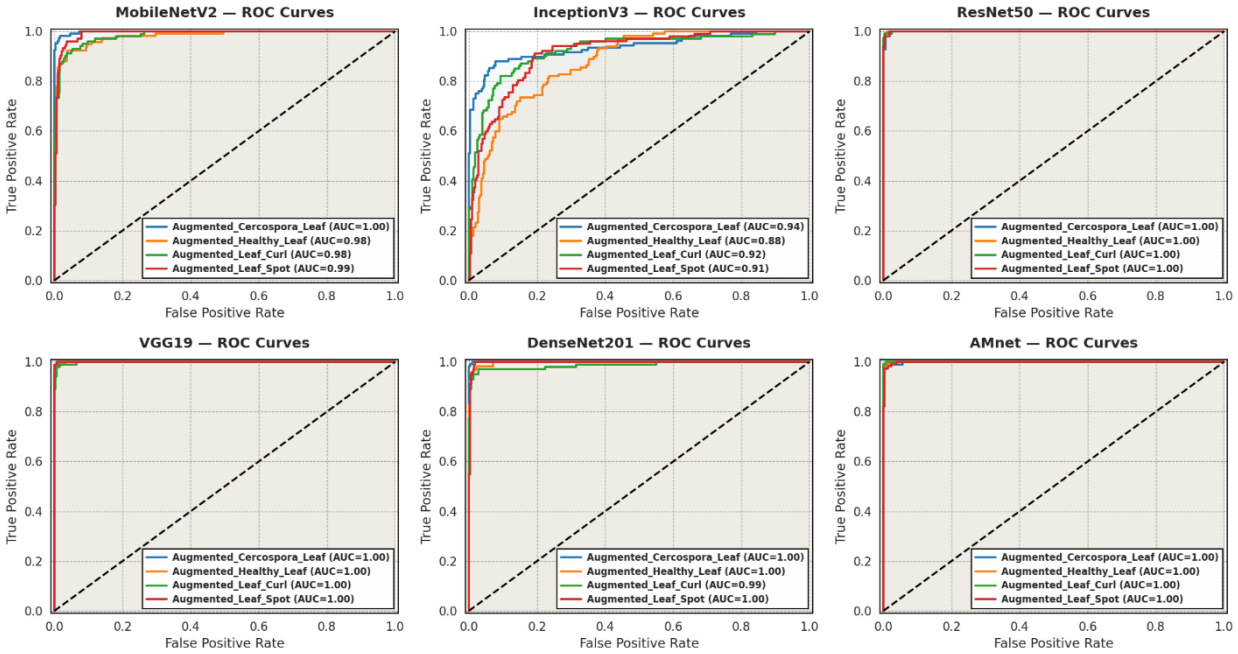


Fig. 6. Models ROC Curves

To examine performance under varying decision thresholds, the precision–recall curves are presented in **Fig. 7**. AMnet, ResNet50, VGG19, and DenseNet201 maintained high precision across most recall levels, with average precision values close to 1.00. InceptionV3 showed noticeably weaker curves, particularly for healthy leaves and leaf-spot classes. This difference suggests that the stronger models learned more discriminative class features, enabling them to retain high recall while limiting false-positive predictions.

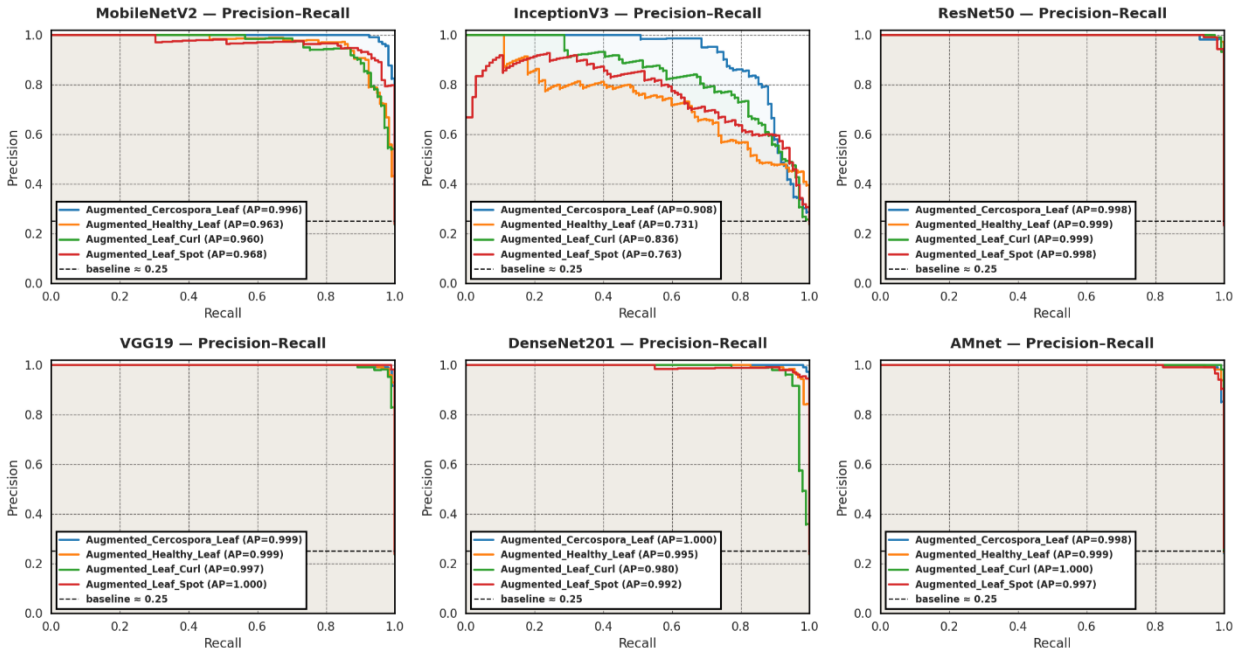


Fig. 7. Models Precision-Recall Curves

4.6 Explainable AI (XAI) Outcomes of AMnet

The explanation results in **Fig. 8** show which parts of the images influenced AMnet's predictions. Grad-CAM highlights wider areas of the leaf, while LIME points to smaller regions that were important for each decision. Most highlighted areas match useful visual features, such as leaf colour, surface texture, veins, and disease spots. This suggests that AMnet mainly focused on relevant leaf regions instead of the background, which makes its predictions easier to understand and more trustworthy.

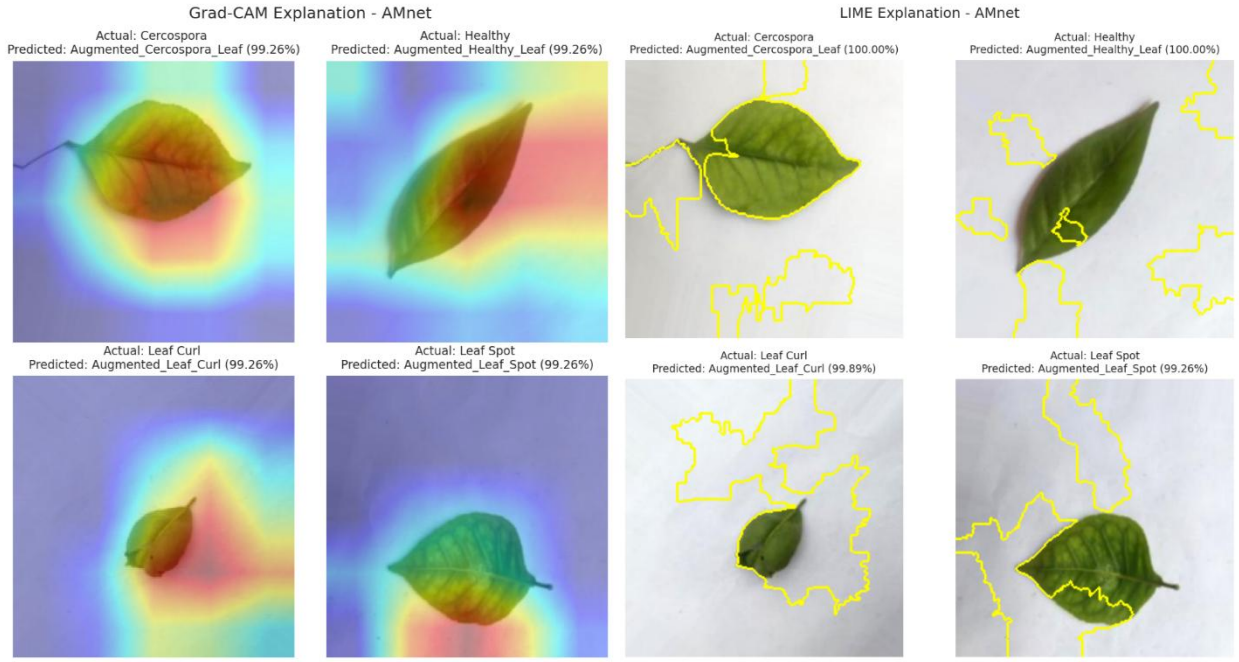


Fig. 8. AMnet XAI Outcome

Across 40 samples, AMnet achieved an average classification accuracy of 87.50%, correctly classifying 35 cases. The quantitative explanation results are summarized in **Table 7**. LIME produced slightly higher mean scores for insertion AUC, deletion AUC, and faithfulness than

Grad-CAM. However, all p-values were above 0.05, indicating that the differences were not statistically significant. This similarity may occur because both methods identified comparable image regions, although they explain model decisions using different approaches.

Table 7. Quantitative comparison of Grad-CAM and LIME explanation method

Metric	Grad-CAM Mean ± SD	LIME Mean ± SD	Mean Difference	Wilcoxon Statistic	p-value
Insertion AUC	0.8888 ± 0.1622	0.8982 ± 0.1524	-0.0095	297.0000	0.1316
Deletion AUC	0.4956 ± 0.3318	0.5052 ± 0.3622	-0.0097	409.0000	0.9947
Faithfulness	0.1069 ± 0.2212	0.1639 ± 0.2643	-0.0570	298.0000	0.1351

The insertion and deletion curves in **Fig. 9** provide further evidence of explanation quality. LIME increases the target-class probability more quickly at the early insertion stages, while Grad-CAM reaches similar values as more important regions are added. In the deletion test, Grad-CAM produces a slightly larger early reduction in probability, although both methods show similar behaviour overall. This pattern suggests that both approaches identify useful image regions, but they rank and represent these regions differently.

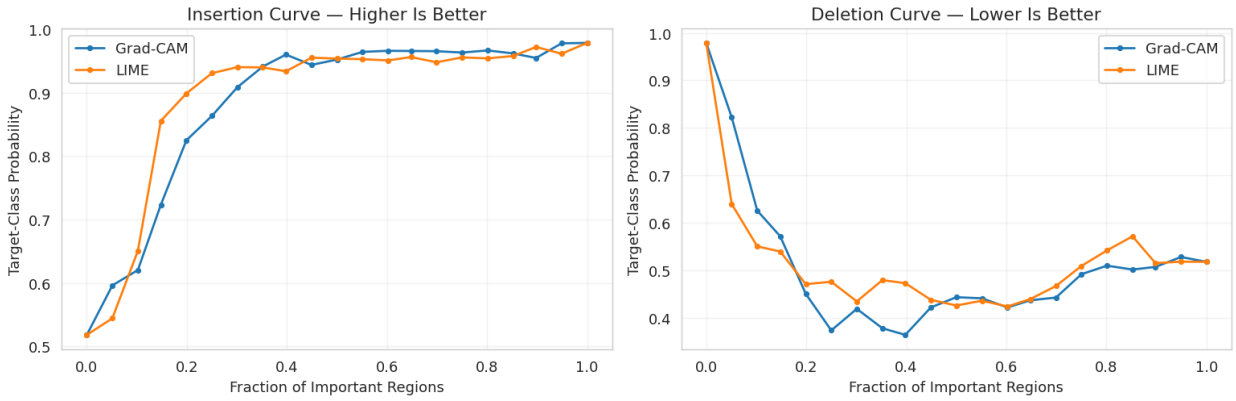


Fig. 9. Insertion and deletion curves for the Grad-CAM and LIME explanations

4.7 Principal Component Analysis of AMnet

Fig. 10 illustrates the elbow-method analysis for selecting the optimal number of clusters using K-means inertia. The inertia decreases sharply from K = 2 to K = 4, indicating a substantial improvement in within-cluster compactness as the number of clusters increases. After K = 4, the reduction in inertia becomes gradual, suggesting limited additional benefit from adding more clusters. Therefore, K = 4 can be considered the most appropriate cluster number, which is also consistent with the four target leaf classes used in this study.

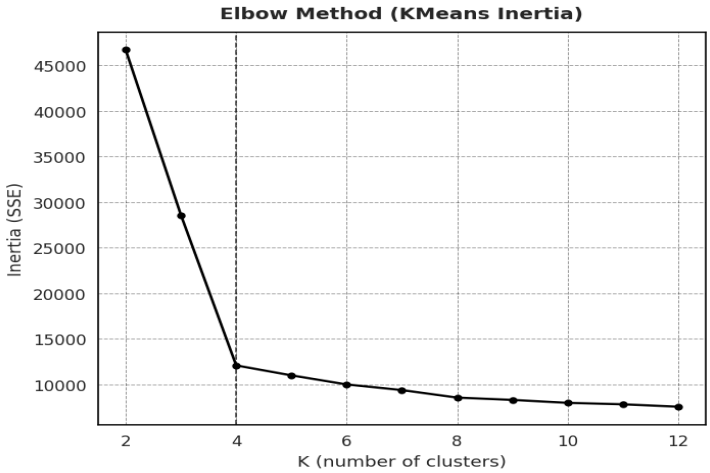


Fig. 10. Elbow Method Plot

The clustering evaluation in **Fig. 11** supports K = 4 as the optimal cluster setting, with the highest Silhouette score (0.597) and the lowest Davies–Bouldin index (0.568). This indicates that the PCA-reduced feature space forms the most compact and well-separated clusters when four clusters are used. The result is also consistent with the dataset structure, since the classification task contains four leaf classes. Therefore, the clustering analysis confirms that the learned feature representation preserves meaningful class-level separability.

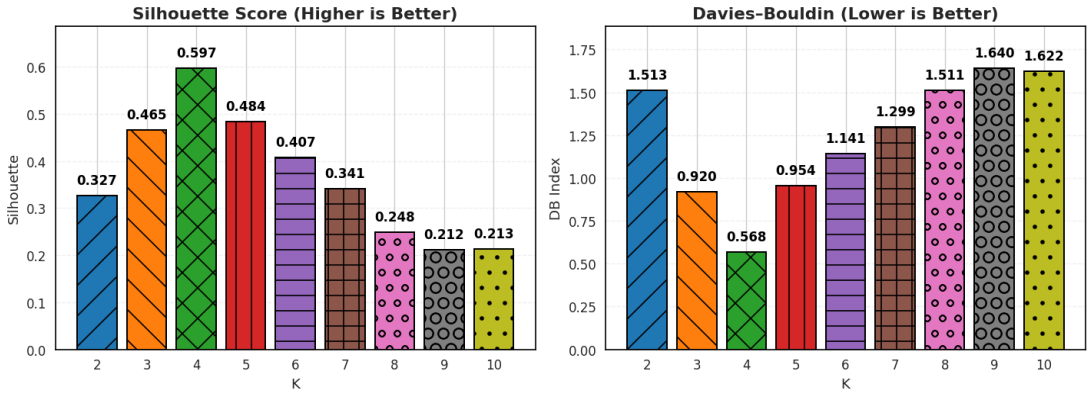


Fig. 11. Clustering Evaluation Metrics for PCA-Reduced Feature Space

A clear class structure is visible in **Fig. 12**, where the learned features form four main groups. Cercospora, leaf curl, and leaf spot samples are mostly separated, while healthy leaves show limited overlap with nearby classes. This separation suggests that AMnet learned useful differences in colour, texture, shape, and disease patterns. The small overlap may result from similar visual features among some leaves and from reducing high-dimensional features to only two principal components.

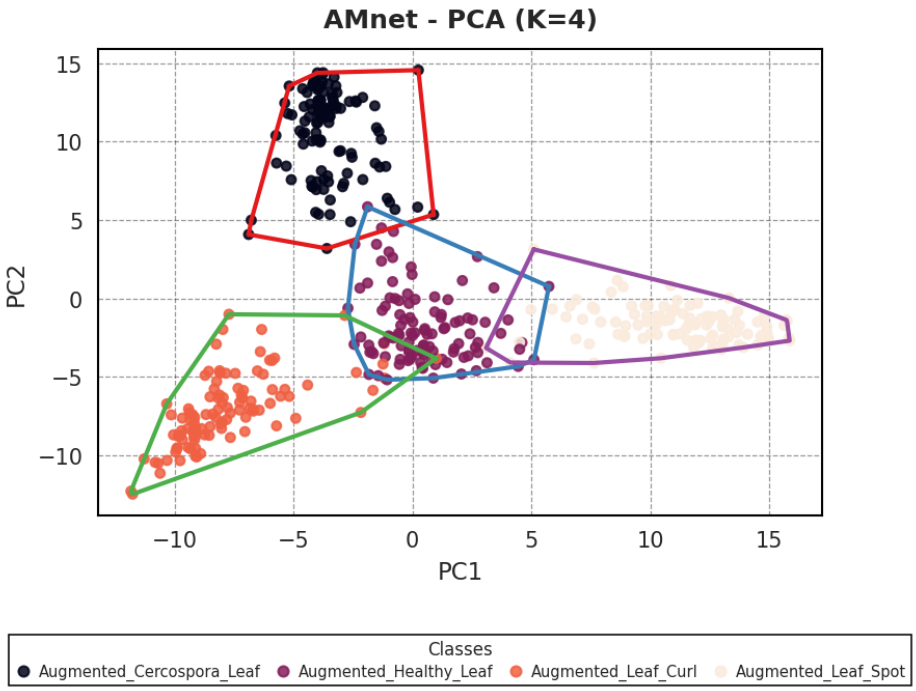


Fig. 12. PCA Visualization of Learned Feature Embeddings from AMnet

Fig. 13 presents the cluster composition of AMnet feature embeddings, showing that each cluster is dominated by a single leaf class. Cluster 0 mainly represents healthy leaf samples (98.1%), Cluster 1 perfectly corresponds to leaf curl (100.0%), Cluster 2 is dominated by leaf spot (97.3%), and Cluster 3 mainly contains Cercospora leaf samples (99.0%). The very low cross-class contamination indicates strong intra-class compactness and inter-class separation in the learned feature space. This result further supports the PCA and clustering analyses, confirming that AMnet learns discriminative and class-consistent representations.

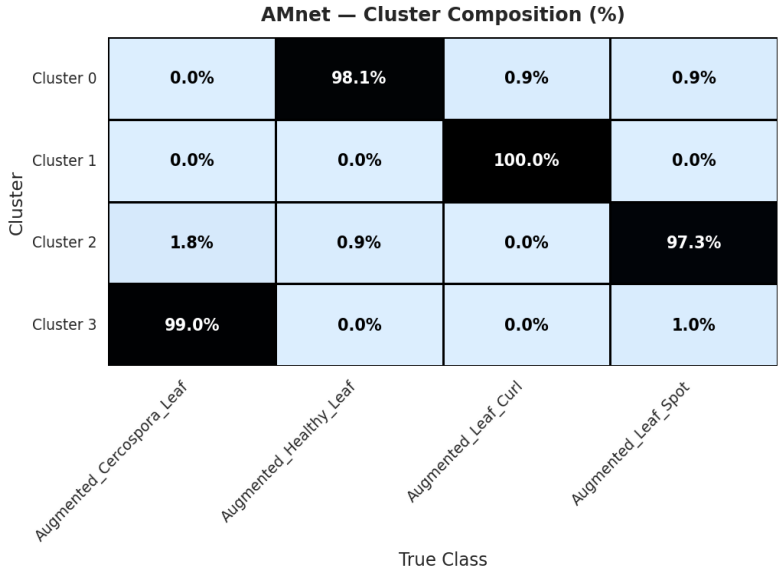


Fig. 13. Cluster Composition (%)

Fig. 14 shows the class centroid distance matrix of AMnet embeddings, where all off-diagonal distances are high, ranging from 17.42 to 18.93. The largest separation is observed between leaf curl and leaf spot (18.93), while the smallest distance occurs between Cercospora leaf and leaf spot (17.42), suggesting relatively closer feature similarity between these two disease classes. The

consistently large centroid distances indicate strong inter-class separation in the learned feature space. This further supports the clustering and PCA results, confirming that AMnet generates discriminative class-level representations.

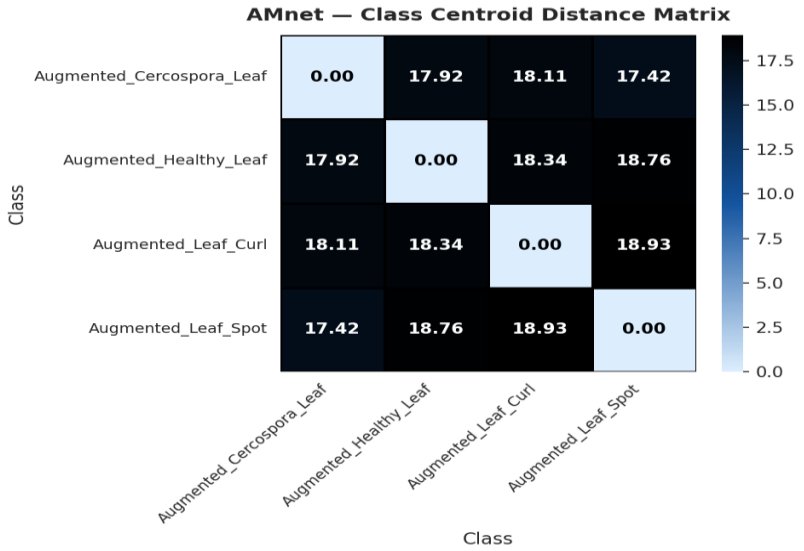


Fig. 14. Class Centroid Distance Matrix

4.8 Statistical Analysis of AMnet

To examine the reliability of AMnet, confidence intervals were calculated for its main performance measures. As reported in **Table 8**, the model achieved 98.36% accuracy, 98.37% macro F1 score, and 97.81% MCC. The lower bounds of all 95% confidence intervals remained above 95%, showing that the model maintained strong performance across repeated samples. The relatively narrow intervals also suggest limited uncertainty, which may result from consistent predictions across the four classes. Overall, these findings support the statistical stability and reliability of AMnet.

Table 8. Overall Performance Metrics with 95 % Confidence Intervals

Metric	Value	95 % CI (Wilson / Bootstrap)
Accuracy	98.36%	Wilson CI: 96.66% – 99.21%; Bootstrap CI: 97.20% – 99.53%
Macro F1	98.37%	Bootstrap CI: 96.99% – 99.53%
MCC	97.81%	Bootstrap CI: 95.99% – 99.08%

4.9 Hyperparameter Sensitivity Analysis of AMnet

To understand how different design choices affect AMnet, several controlled model variants were evaluated. As summarized in **Table 9**, the study focused on changes to the GCN hidden dimension and dropout rate while keeping the main InceptionV3–GCN structure unchanged. This approach was selected because AMnet is built around a fixed graph-based architecture; therefore, the present ablation mainly examines hyperparameter sensitivity rather than removing the GCN entirely. Variants A2 and A4 test whether weaker or stronger regularization improves generalization, while A3 examines whether a smaller GCN can reduce model complexity without losing important relational information. These comparisons help identify a suitable balance between learning capacity, overfitting control, and computational efficiency.

Table 9. AMnet variants used for hyperparameter-based ablation analysis

Model ID	Variant Name	Key Modification	Expected Outcome
A1	InceptionV3 and GCN	node_dim=512, gcn_hidden=256, dropout=0.3, grid_size=5	Tests whether spatial graph reasoning improves over the plain CNN baseline.
A2	InceptionV3 and GCN with LowDropout	Only dropout is reduced from 0.3 to 0.2	Tests whether lower dropout improves accuracy by reducing underfitting.
A3	InceptionV3 with LightGCN	GCN hidden dimension is reduced from 256 to 128	Tests whether a smaller GCN can reduce overfitting and computation.
A4	InceptionV3 and GCN HighDropout	Only dropout is increased from 0.3 to 0.4	Tests whether stronger dropout improves generalization.

The effect of each hyperparameter setting is presented in **Table 10**. Among the four variants, A4 achieved the best overall performance, with 98.83% accuracy, 98.84% F1 score, 99.95% PR AUC, and 98.45% MCC. A2 also performed strongly, whereas A1 and A3 produced lower results. The improvement in A4 suggests that increasing dropout to 0.4 provided stronger regularization and reduced overfitting. In contrast, reducing the GCN hidden dimension in A3 may have limited the model's ability to learn complex feature relationships. Overall, A4 offered the best balance between model capacity and generalization.

Table 10. Performance comparison of model variants in the ablation study across multiple evaluation metrics

Model ID	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	PR AUC (%)	MCC (%)
A1	95.09	95.64	95.21	95.23	99.69	93.60
A2	98.36	98.40	98.36	98.37	99.83	97.82
A3	95.79	96.04	95.94	95.79	99.72	94.53
A4	**98.83**	**98.86**	**98.85**	**98.84**	**99.95**	**98.45**

4.10 Model Performance on Four-Fold Cross-Validation

To evaluate whether AMnet performs consistently across different data splits, four-fold cross-validation was conducted. As reported in **Table 11**, the model achieved mean accuracy, precision, recall, and F1 score values of approximately 97.35%. The very small standard deviations indicate that performance changed only slightly between folds. This stability suggests that AMnet did not depend heavily on a particular training–testing split and was able to learn features that remained useful across different subsets of the dataset. Therefore, the cross-validation results provide further evidence of the model's reliability and generalization ability.

Table 11. Four-fold cross-validation performance of AMnet

Model Name	Mean Results (%)	Standard Deviation
Accuracy	97.35	0.0024
Precision	97.36	0.0023
Recall	97.36	0.0025
F1 Score	97.35	0.0024

4.11 Model Performance on Additional Dataset

To examine whether the models remained effective beyond the main dataset, an additional groundnut leaf dataset developed by Manvikar and Reddy was used as a separate benchmark [40]. The dataset contains 10,361 images collected under natural conditions in Koppal, Karnataka, India, with support from a plant pathologist. It includes six classes: healthy leaves, early leaf spot, late leaf spot, nutritional deficiency, rust, and early rust. This dataset was selected because its independent source, field-based images, and different disease categories provide a practical test of model robustness and reduce the possibility that performance is limited to the original dataset.

The results in **Table 12** show that AMnet achieved the highest accuracy (81.25%), recall (80.30%), F1 score (80.11%), PR AUC (87.51%), and MCC (76.55%). DenseNet201 achieved the highest precision at 81.41%, but its other scores were lower. The reduced performance compared with the main dataset may be caused by differences in image conditions, disease appearance, and class structure. Nevertheless, AMnet maintained the strongest overall performance, indicating better adaptability to a different dataset.

Table 12. Performance of the evaluated models on the additional groundnut leaf dataset

Model Name	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	PR AUC (%)	MCC (%)
MobileNetV2	68.50	70.46	68.38	67.43	76.09	61.41
VGG19	77.50	77.97	77.38	77.04	86.10	72.18
DenseNet201	77.25	81.41	76.88	75.71	85.17	73.00
AMnet	**81.25**	**80.37**	**80.30**	**80.11**	**87.51**	**76.55**

4.12 Prototype Application

To demonstrate the practical use of AMnet, a prototype application was developed using Gradio, as shown in **Fig. 15**. The interface allows users to upload a leaf image and obtain the predicted class together with class probabilities. It also displays a Grad-CAM heatmap to show the image regions that influenced the prediction. This design makes the model easier to use and understand, while providing visual support for its decisions. The prototype serves as an initial step toward a user-friendly disease screening tool.

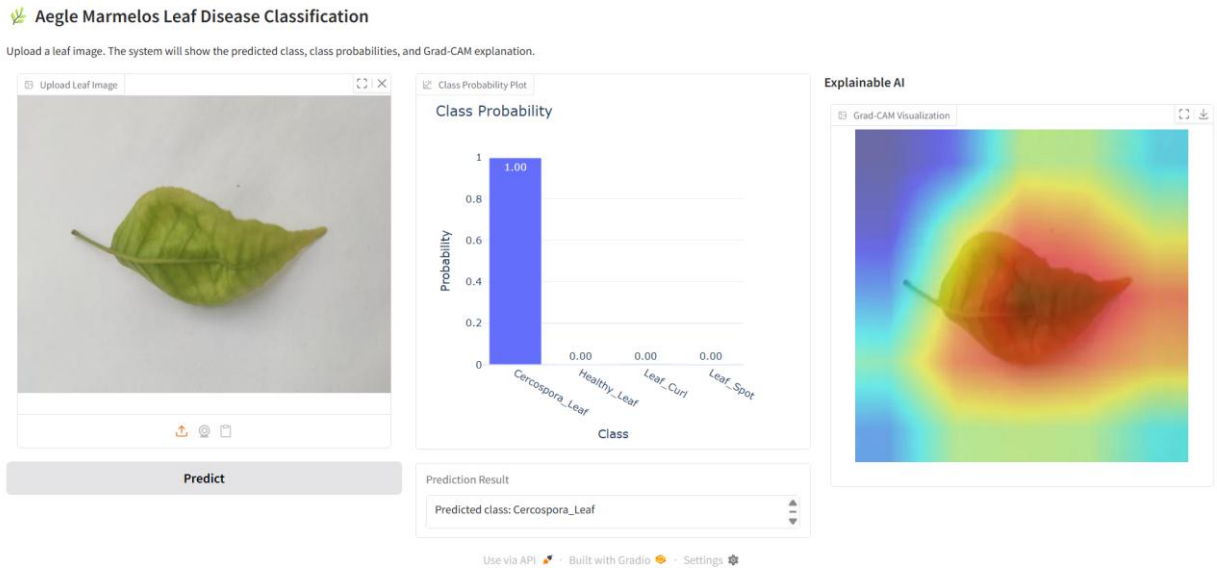


Fig. 15. Prototype Gradio Application

5 Discussion

5.1 Principal Findings

AMnet achieved 98.83% accuracy, 98.84% F1-score, 99.95% PR-AUC, and 98.45% MCC for four-class *Aegle marmelos* leaf classification, outperforming the evaluated CNN baselines. Confidence intervals and four-fold cross-validation further indicated stable performance across different data partitions. The improvement is likely associated with the combination of convolutional and graph-based feature learning. While InceptionV3 extracts local patterns such as lesions, color variation, venation, and deformation, the fixed GCN models relationships among neighboring regions in a 5×5 feature grid. Because foliar symptoms often extend across adjacent areas, this spatial reasoning may provide more informative representations than conventional CNN features alone. However, the graph captures spatial associations within an image rather than the biological progression of disease.

5.2 Failure Cases and Model Reliability

AMnet maintained balanced performance across the four classes, and the PCA-based analysis revealed comparatively compact, well-separated feature groups. Nevertheless, occasional confusion occurred between Cercospora and leaf spot. These classes may share irregular lesions, discoloration, and similar surface textures, particularly at mild or early disease stages. Variations in illumination, image scale, orientation, and symptom severity may further reduce their visual distinction. More expert-confirmed samples representing different disease stages, along with representative misclassified images, are therefore needed. Uncertainty estimation could also help identify ambiguous cases requiring specialist review.

5.3 Explainability and Biological Relevance

Grad-CAM and LIME generally highlighted lesions, discoloration, venation, texture changes, and deformed leaf regions rather than only the background. The quantitative explanation analysis also suggested that these regions contributed to the predictions.

However, the highlighted areas were not independently verified by plant pathologists. The explanations should therefore be considered visually plausible rather than biologically confirmed. Future work should compare explanation maps with expert-annotated disease regions to establish their diagnostic relevance.

5.4 Practical Deployment

The Gradio prototype demonstrates AMnet's practical usability in a web-based environment, while the reported inference time suggests potential for rapid prediction. However, the model has not been evaluated on mobile or embedded devices. Its parameter counts and storage requirements may also restrict deployment on resource-constrained platforms. Moreover, the dataset mainly contains images acquired under controlled or semi-controlled conditions. Robustness to complex backgrounds, shadows, variable illumination, occlusion, motion blur, low-resolution images, mixed infections, and unseen severity levels remains untested. Quantization, pruning, lightweight backbones, field-based datasets, and device-specific testing are required before real-world deployment.

5.5 Limitations and Future Work

The main limitation is the use of 1,513 raw images from a single crop and four categories. Although augmentation increased the training data, it did not introduce the biological diversity of independently collected samples. Consequently, generalization to other crops, diseases, locations, and imaging conditions remains uncertain. Future studies should use larger, multi-location, multi-season, and expert-verified datasets. Adaptive graph construction, multiscale reasoning,

uncertainty calibration, pathologist-supported validation, and embedded-device evaluation should also be investigated to establish the broader reliability and practical value of AMnet.

6 Conclusion

This study introduces AMnet, an InceptionV3-based fixed-graph convolutional framework that excels at four-class Aegle marmelos leaf classification. AMnet achieved remarkable results with 98.83% accuracy, 98.84% F1-score, 99.95% PR-AUC, and 98.45% MCC, significantly surpassing the evaluated CNN baselines under the specified experimental conditions. The ablation study clearly identified the high-dropout configuration as optimal for balancing model capacity and generalization. Confidence intervals and four-fold cross-validation robustly confirm the internal reliability of these results. Feature-space analyses unequivocally demonstrated distinct class separability, while Grad-CAM and LIME provided visually compelling explanations of the regions that influence model predictions. The Gradio prototype further showcased AMnet's potential as an accessible decision-support tool. Despite these achievements, the study is constrained by the relatively small single-crop dataset, controlled imaging conditions, the absence of plant pathologist validation, and the lack of embedded-device testing. Future research should prioritize larger field-based datasets, external cross-crop validation, expert evaluation of explanations, adaptive graph construction, and lightweight deployment. In summary, AMnet establishes a promising and reproducible foundation for explainable graph-based leaf disease classification.

Data availability

The dataset analysed of this study is publicly available in the Mendeley Data repository at (https://data.mendeley.com/datasets/54r883j5zr/1).

Code availability

The source code for the GCN-based architecture developed in this study is publicly available in the GitHub repository (https://github.com/RAIYAN8018/Aegle-Marmelos-Model-Architecture).

Declarations

Competing interests

The authors declare no competing interests.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this manuscript, the authors utilized AI-assisted tools to enhance language clarity and readability. All generated or refined content was carefully reviewed and edited by the authors, who take full responsibility for the accuracy, integrity, and originality of the work.

References

[1] Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. Frontiers in Plant Science, 7, 1419. https://doi.org/10.3389/fpls.2016.01419.

[2] Ferentinos, K. P. (2018). Deep learning models for plant disease detection and diagnosis. Computers and Electronics in Agriculture, 145, 311–318. https://doi.org/10.1016/j.compag.2018.01.009.
[3] Barbedo, J. G. A. (2018). Factors influencing the use of deep learning for plant disease recognition. Biosystems Engineering, 172, 84–91. https://doi.org/10.1016/j.biosystemseng.2018.05.013.
[4] Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. ICLR (Poster) 2017
[5] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. Proceedings of the IEEE International Conference on Computer Vision, 618–626. https://doi.org/10.1109/ICCV.2017.74.
[6] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations, 97–101. https://doi.org/10.18653/v1/N16-3020.
[7] Deepika Lakshmi Kanumuri, Madhuri Kamma, Somitha Anna, and Pavan Kumar Kolluru (2025). Attention-based deep learning model for robust pneumonia classification and categorization using image processing. In Proceedings of the ICITSM Part I. EAI. https://doi.org/10.4108/eai.28-4-2025.2357765
[8] Gao, X., Zhang, F., Guo, X. et al. Attention-based deep learning for accurate cell image analysis. Sci Rep 15, 1265 (2025). https://doi.org/10.1038/s41598-025-85608-9
[9] Miao, M., Yan, W., Gao, X., Yang, L., Zhou, J., & Zhang, W. (2025). Attention-Based CNN-BiGRU-Transformer Model for Human Activity Recognition. Applied Sciences, 15(23), 12592. https://doi.org/10.3390/app152312592
[10] N, A., & S, U. (2025). SkinProNet: An attention-based deep learning system for skin disease classification and segmentation. Softw. Impacts, 26, 100798.
[11] Gupta, R., Kumar, V., & Tripathi, V. ESAM: Efficient Spatial Attention based Deep Learning Approach for Fundus Images Classification. Biomedical & Pharmacology Journal 2025;18(4).
[12] Yousafzai, S.N., Nasir, I.M., Mansour, S. et al. A hybrid deep learning approach integrating CNN and transformer for lung cancer classification using CT scans. Sci Rep (2026). https://doi.org/10.1038/s41598-026-41161-7
[13] Ayesha, N., & Khalidi, H. (2026). A Vision Transformer with a Self-Attention Mechanism for High-Accuracy Blood Cell Classification. Engineering, Technology & Applied Science Research, 16(1), 32557–32563. https://doi.org/10.48084/etasr.16600
[14] Fuqiang Kuang. 2026. A CNN-Transformer Hybrid Model Based on Tri-domain Attention for Music Genre Classification. In Proceedings of the 2025 5th International Conference on Big Data, Artificial Intelligence and Risk Management (ICBAR '25). ACM, New York, NY, USA, 378–385. https://doi.org/10.1145/3800227.3800285
[15] Aboelenin, S., Elbasheer, F.A., Eltoukhy, M.M. et al. A hybrid Framework for plant leaf disease detection and classification using convolutional neural networks and vision transformer. Complex Intell. Syst. 11, 142 (2025). https://doi.org/10.1007/s40747-024-01764-x
[16] D, A., T, L. N., & Prusty, M. R. (2026). An attention-augmented lightweight convolutional framework for fine-grained plant leaf disease classification. Frontiers in plant science, 17, 1762956. https://doi.org/10.3389/fpls.2026.1762956
[17] Sandhi, A., Kumar, R., Kumar, D., & Bhardwaj, R. (2025). Multi-architecture CNN framework with MBConv-SE optimization for fruit disease classification. Applied Fruit Science, 67(4), 277.

[18] Sandhi, A., Kumar, R., Bhardwaj, R., & Kumar, D. (2026). SAAEffNet: A Synthetic-Augmented Attention EfficientNet Model for Mango Disease Classification. Applied Fruit Science, 68(2), 93.
[19] Ilsever, M., & Baz, I. (2024). Consistency Regularization Based Semi-supervised Plant Disease Recognition. Smart Agricultural Technology, Volume 9, 2024, 100613, ISSN 2772-3755, https://doi.org/10.1016/j.atech.2024.100613 .
[20] Pham, TC., Nguyen, TN. & Nguyen, VD. Ambiguity-aware semi-supervised learning for leaf disease classification. Sci Rep 15, 14070 (2025). https://doi.org/10.1038/s41598-025-95849-3
[21] Qiu, Y., Meng, J., & Li, B. (2025). Semi-supervised Strong-Teacher Consistency Learning for few-shot cardiac MRI image segmentation. Computer methods and programs in biomedicine, 261, 108613. https://doi.org/10.1016/j.cmpb.2025.108613
[22] Huang, S., Wang, S., Qiao, S., Liu, Y., & Liu, T. (2025). SEMI-PLC: A framework for semi-supervised medical images segmentation with pseudo label correction. Computer methods and programs in biomedicine, 271, 109027. https://doi.org/10.1016/j.cmpb.2025.109027
[23] Rahmati, B., Shirani, S. & Keshavarz-Motamed, Z. A hybrid approach for enhancing pseudo-labeling in medical images through pseudo-label refinement. Sci Rep 15, 35161 (2025). https://doi.org/10.1038/s41598-025-19121-4
[24] Faysal, A., Rostami, M., Roshan, R. G., Muralidhar, N., & Wang, H. (2026). Semi-Supervised Masked Autoencoders: Unlocking Vision Transformer Potential with Limited Data. ArXiv.org. https://arxiv.org/abs/2601.20072
[25] B. Islam et al., "Annotation-Efficient Lung Cancer Classification: A Comparative Study of Supervised, Self-Supervised, and Semi-Supervised Learning," 2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON), Dhaka, Bangladesh, 2025, pp. 835-840, doi: 10.1109/BECITHCON69222.2025.11504014 .
[26] Singh, B., Sharma, R. P., & Dey, S. (2025). Interpretable Plant Leaf Disease Detection Using Attention-Enhanced CNN. ArXiv.org. https://arxiv.org/abs/2512.17864
[27] Sandhi, A., Kumar, R., Bhardwaj, R., Kumar, D., Rana, A. K., Ajala, O., ... & Salau, A. O. (2025). Optimized deep learning framework for pomegranate disease detection using nature-inspired algorithms. Plant Methods, 21(1), 124.
[28] Sanyal, M., Dasgupta, S., Chaki, R. et al. Pseudo CNN: An Updated CNN Based Semi Supervised Learning Model for Leaf Disease Detection. Innovations Syst Softw Eng 21, 1511–1520 (2025). https://doi.org/10.1007/s11334-025-00617-1
[29] Xu, H., & Zhang, X. (2025). Graph Convolutional Network with Agent Attention for Recognizing Digital Ink Chinese Characters Written by International Students. Information, 16(9), 729. https://doi.org/10.3390/info16090729
[30] Yan, P., An, T., Yu, J. et al. Graph convolutional network with reinforced dependency graph and denoising mechanism for sarcasm detection. Sci Rep 15, 35094 (2025). https://doi.org/10.1038/s41598-025-18849-3
[31] Thekumparampil, Kiran K, Wang, C., Oh, S., & Li, L.-J. (2018). Attention-based Graph Neural Network for Semi-supervised Learning. ArXiv.org. https://arxiv.org/abs/1803.03735
[32] Yang, G., Li, Y., Zhou, W., Wu, Y., Wu, W., Gu, X. (2022). Attention Mechanism Based on Temporal Graph Convolutional Neural Network for Traffic Flow Prediction. In: Jia, Y., Zhang, W., Fu, Y., Yu, Z., Zheng, S. (eds) Proceedings of 2021 Chinese Intelligent Systems Conference. Lecture Notes in Electrical Engineering, vol 804. Springer, Singapore. https://doi.org/10.1007/978-981-16-6324-6_44

[33] Su, L., Huang, J., & Li, F. (2026). Adaptive graph convolution neural networks with spatial and frequency domain attention mechanism for traffic flow forecasting. Applied Soft Computing, 197, 115156. https://doi.org/10.1016/j.asoc.2026.115156
[34] Luo, X., Zhu, C., Zhang, D., & Li, Q. (2023). Dynamic Graph Convolutional Network with Attention Fusion for Traffic Flow Prediction. ArXiv.org. https://arxiv.org/abs/2302.12598
[35] Yang, H., Jiang, C., Song, Y. et al. TARGCN: temporal attention recurrent graph convolutional neural network for traffic prediction. Complex Intell. Syst. 10, 8179–8196 (2024). https://doi.org/10.1007/s40747-024-01601-1
[36] Chang, Q., Li, X., & Duan, Z. (2024). Graph global attention network with memory: A deep learning approach for fake news detection. Neural Networks, 172, Article 106115. https://doi.org/10.1016/j.neunet.2024.106115
[37] Surekha, B., Subha Mastan Rao, T. AdjLeafGNN: a hybrid deep learning and graph neural network framework for probabilistic modeling of adjacent leaf disease spread in precision agriculture. Sci Rep (2026). https://doi.org/10.1038/s41598-026-45061-8
[38] Sundhar S, Sharma R, Maheshwari P, Kumar SR and Kumar TS (2025) Enhancing leaf disease classification using GAT-GCN hybrid model. Front. Plant Sci. 16:1569821. doi: 10.3389/fpls.2025.1569821
[39] Huq, Rezwan; Ahmed, Jubaer; Nessa, Maherun; Islam, Tasmia; Gani, Raiyan (2024), "Smartphone Image Dataset for Aegle Marmelos, Hog Plum and Lemon Plant Leaf ", Mendeley Data, V1, doi: 10.17632/54r883j5zr.1
[40] Manvikar, Aishwarya; Reddy, Padmanabha (2023), "Dataset of groundnut plant leaf images for classification and detection", Mendeley Data, V3, doi: 10.17632/22p2vcbxfk.3